

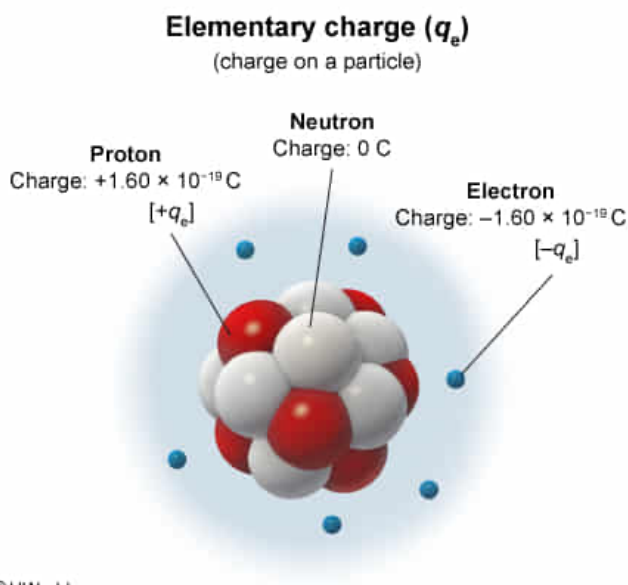
A scientist observes that electric charge is scattered in random directions after a solid mass is exposed to isolated pulses of electromagnetic radiation. As the scientist measures the charge that gradually accumulates on detectors that surround the solid mass, the quantity of charge increases only in multiples of $1.6 \times 10^{-19} \text{ C}$ because:

- ☐ A. no electric charge larger than $1.6 \times 10^{-19} \text{ C}$ has been isolated in nature [1%]
- ✓ ☒ B. $1.6 \times 10^{-19} \text{ C}$ is the magnitude of elementary charge [84%]
- ☐ C. neutrons with a charge of $1.6 \times 10^{-19} \text{ C}$ are ejected from the solid mass [3%]
- ☐ D. only particles with a charge of $1.6 \times 10^{-19} \text{ C}$ are small enough to acquire the velocity sufficient to reach the detectors [10%]

Correct

84% answered correctly

Explanation:



Electric charge is a fundamental property of matter that influences many phenomena. Several conventions exist by which the magnitude of charge is quantified. Many of these conventions relate to the basic structure of the atom, which consists of positively charged **protons**, uncharged **neutrons**, and negatively charged **electrons**.

The **SI unit** of electric charge is the coulomb (C), where +1 C is equivalent in

charge to approximately 6.24×10^{18} protons, and -1 C is equivalent in charge to approximately 6.24×10^{18} electrons:

$$1 \text{ C} = 6.24 \times 10^{18} \text{ proton charges} \quad -1 \text{ C} = 6.24 \times 10^{18} \text{ electron charges}$$

The charge magnitudes of one electron and one proton are not only equal but also the smallest unit of charge known as the **elementary charge q_e** :

$$\frac{+1 \text{ C}}{6.24 \times 10^{18}} = 1 \text{ proton charge} \quad \frac{-1 \text{ C}}{6.24 \times 10^{18}} = 1 \text{ electron charge}$$
$$+1.60 \times 10^{-19} \text{ C} = +1 q_e \quad -1.60 \times 10^{-19} \text{ C} = -1 q_e$$

In this question, a scientist observes that exposing a solid mass to electromagnetic radiation causes charge to accumulate in multiples of $1.6 \times 10^{-19} \text{ C}$, or q_e . Therefore, the likely explanation for this observation is that electromagnetic radiation is causing protons or electrons to be ejected from the atoms of the solid mass.

(Choice A) No electric charge *smaller* than $1.6 \times 10^{-19} \text{ C}$ has been isolated in nature, although magnitudes of charge greater than $1.6 \times 10^{-19} \text{ C}$ are frequently observed.

(Choice C) As their name suggests, neutrons are neutral, uncharged elementary particles that are found within the nucleus of the atom.

(Choice D) Charge is a separate physical quantity from mass and therefore has no direct influence on the translational motion of that object in the absence of a durable electromagnetic field.

Educational objective:

The elementary charge is the smallest unit of charge observed in nature, equal to $1.6 \times 10^{-19} \text{ C}$. The elementary charge is equal in magnitude and sign to the charge of a proton, and equal in magnitude but opposite in sign to the charge of an electron.

Time spent: 0 sec

Subject:
Physics

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System:
Electrostatics and Circuits